Hence,

Number of coils = Total number of slots

9.6.14 Balanced winding

If the number of coils in all coil groups per phase are equal, then this type of winding is referred to as balanced winding.

9.6.15 Unbalanced winding

If the number of coils in all coil groups per phase are unequal, it is called unbalanced winding.

Do you Know?



Metal-Air batteries are small in size and have light weight batteries.

But its ranges are high. When not in use, they quickly disrupt.

Because metal electrodes are affected by corrosion.

9.6.16 Whole coil winding

If the number of coil groups per phase is equal to the number of poles, then the winding is called as whole coil winding. All the three phase lap windings and single phase concentric windings belong to this type.

9.6.17 Half coil winding

If the number of coil groups per phase is equal to the half of the number of poles, then the winding is called as half coil winding. This type of winding is used in three phase concentric winding.

9.6.18 Concentrated winding

If all the conductors of one coil side are inserted in single slot for making one complete pole or if we place a single coil for making one complete pole, it is called as concentrated winding.

Three phase half coil winding, field winding on pole of DC & AC generators are the examples of concentrated winding.

9.6.19 Distributed winding

Here the coils for a pole should not be made as a single coil and hence the pole contains several coils and inserted in several slots. This type of winding is known as distributed winding. All the stators and rotor windings of AC machines and armature winding of DC machines followed this type of winding.

9.6.20 AC single phase concentric winding

A.C, 1ϕ, 4 poles, 24 slots concentric winding for a split phase capacitor type motor.

$$\text{Pole pitch} = \frac{\textit{Number of slots}}{\textit{Number of poles}}$$

$$= \frac{24}{4}$$

$$= 6$$

Coil pitch = 1 + pole pitch

= 1+6 = 7

=1-7 (full pitched)

1-6 (short pitched)

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